Timing Diagram

A timing diagram in a microprocessor is a graphical representation showing the precise, clock-by-clock sequence of signals (address, data, and control) during instruction execution. It visualizes T-states (clock cycles) to illustrate machine cycles like Opcode Fetch, Memory Read/Write, or I/O operations.

Key Components & Concepts:

- A T-state (time state) in a microprocessor is the smallest, fundamental unit of time, equal to exactly one clock cycle period (1/f). These states constitute the building blocks of machine cycles and instruction cycles.
- A machine cycle in a microprocessor is the fundamental time required to complete a single operation of accessing memory or I/O devices, such as reading or writing data. It is a subset of an instruction cycle and consists of multiple clock periods (T-states), typically 3 to 6, depending on the operation.

  Key Aspects of Machine Cycles

  1- Opcod fetch cycle: **(4T-6T):** The first machine cycle of every instruction, used to fetch the instruction from memory.

  2- Memory Read/Write cycle **(3T)**: Used to read data from or write data to memory.

  3- I/O Read/Write cycle **(3T)**: Used to transfer data between the microprocessor and I/O devices.

  4- **T-States:** Each machine cycle is subdivided into T-states (one T-state = one clock cycle) where specific actions like setting the address bus, enabling ALE, or activating read/write signals occur.

Common 8086 Machine Cycle Types:

1. **Opcode Fetch:** Fetches the instruction code.
2. **Memory Read:** Reads data from memory.

3. **Memory Write:** Writes data to memory.
4. **I/O Read:** Reads data from an input device.
5. **I/O Write:** Writes data to an output device.
6. **Interrupt Acknowledge:** Responds to interrupts.

- **Purpose of Timing Diagrams**:

1- Visualization: Maps the interaction between the CPU and memory /peripherals.

2- Troubleshooting: Helps engineers analyze signal delays and timing violations.

3- Interface Design: Crucial for designing hardware that matches the speed of the microprocessor.

1- Opcod Fetch Cycle

In the 8086 microprocessor, the Opcode Fetch machine cycle is the first cycle of every instruction. It typically lasts 4 T-states (clock periods), though it can extend up to 6 T-states for complex instructions. During this cycle, the processor retrieves the 1-byte Opcode from memory and decodes it to determine the next steps.

Timing Diagram of Opcode cycle breakdown (4 T-states)

T1 : The Program Counter (PC) places the 16-bit address on the buses: Higher-order bits (A_{15}- A_8) on the address bus and lower-order bits (A_7- A_0) on the multiplexed bus. ALE (Address Latch Enable) goes high to signal that the bus contains an address. Status signals are set to identify the Opcode Fetch operation.

T2 : ALE goes low, and the multiplexed bus () is released to serve as a data bus. The (Read) signal goes low, enabling the memory to place the opcode on the data bus.

T3: The processor reads the Opcode from the data bus into the Instruction Register (IR). Once the data is received, the signal goes back high, disabling the memory.

T4 The processor decodes the Opcode and prepares for execution. The address/data buses enter a high-impedance state. Figure in the below shows an example of Opcode timing diagram.

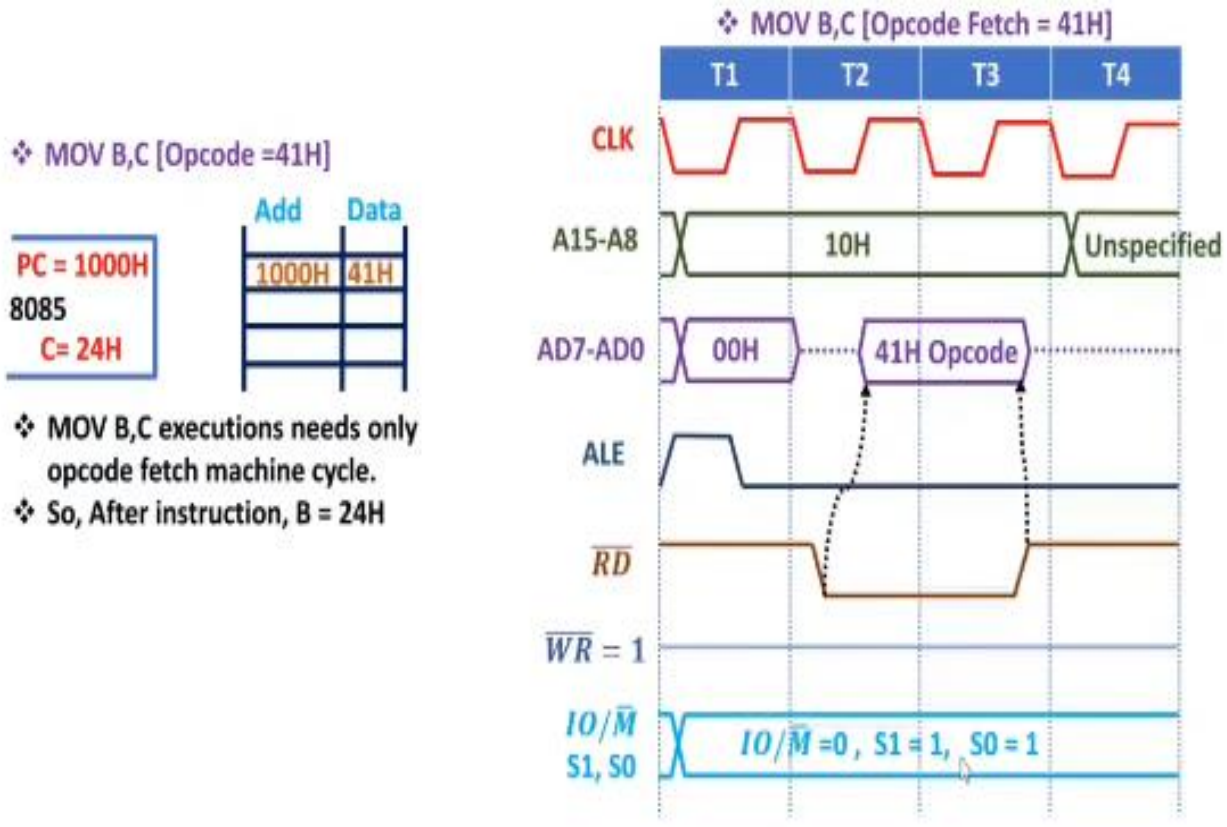


Figure: Opcode Fetch machine cycle

2- Memory read machine cycle

A memory read machine cycle in a microprocessor (like the 8085) is a 3-T-state process used to fetch data or instructions from a specific memory location, such as RAM or ROM. It involves placin$\overline{M}$he address on the bus, activating control signals (IO/), and transferring data to a register.

Key Aspects of Memory Read Machine Cycle:

- **Duration:** Typically 3 T-states (T_1, T_2, T_3)
- **Purpose:** To read data bytes from memory (different from Opcode Fetch, which fetches the first byte of an instruction).
- **Signals:** The IO/M signal is set to low (0) to indicate a memory operation, and the signal goes low (active)

Steps in the Memory Read Cycle:

1. **T_1 State**: The microprocessor places the 16-bit address of the memory location on the Address Bus (A_8-A_{15}) and the Multiplexed Address/Data Bus (AD_0-AD_7). The Address Latch Enable (ALU) signal goes high to latch the lower-order address.
2. **T2 State**: The RD (read) signal goes low to enable the memory device, and the address bus is freed to be used for the data transfer. The memory device puts data on the data bus (AD_0-AD_7).
3. **T3 State**: The microprocessor reads the data from the data bus, stores it in a register, and sets the RD signal to high (inactive), disabling the memory device.

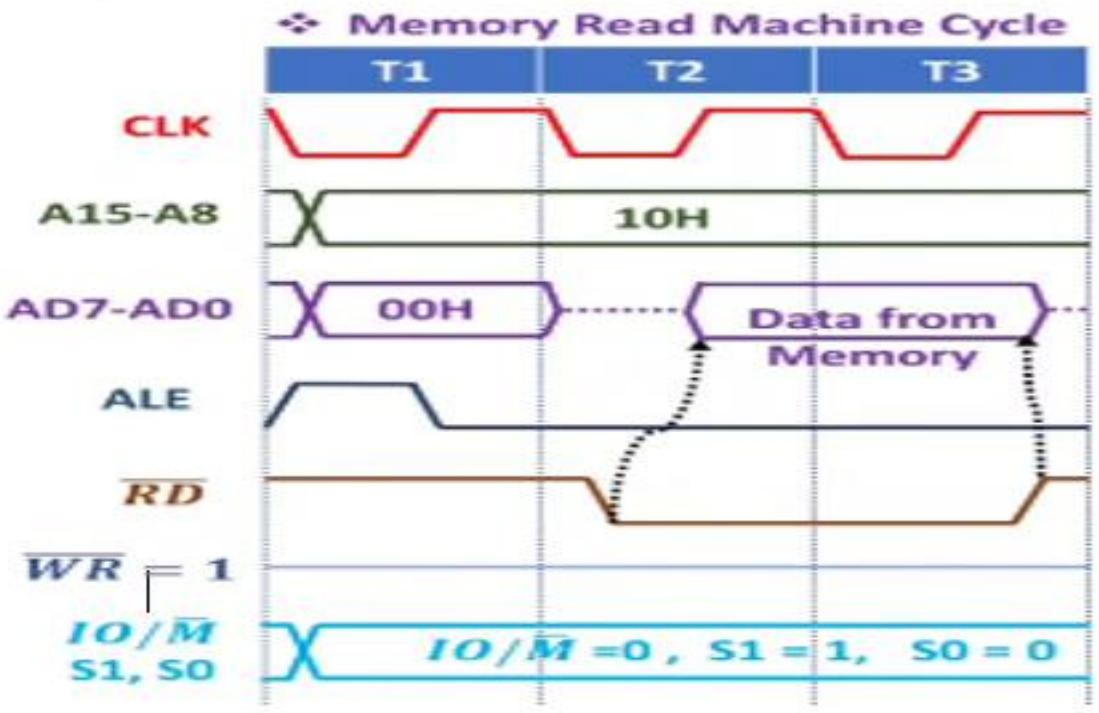


Figure: Memory Read machine cycle

3- Memory write Machine cycle

The microprocessor executes the memory write cycle to store the data into RAM or stack memory. 8085 processor executes this machine cycle in 3 T-states. Steps below show the details of this machine cycle:

1- **Step1 (T1 state):** processor places the address on the address lines from SP or Rp and activates ALE in order to latch low-order of address. Also, it sends the status signals with logical status (0 0 1) for memory write machine cycle.
2- **Step2 (T2 state):** , 8085 processor places that data on data bus and activates the WR signal to writing data into addressed memory location.

3- **Step 3 (T3 state):** The processor deactivates the WR signal which disables the memory device and terminates the write operation.



Figure: Memory Write Machine Cycle